

Grembi Lab Manual

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Chapter 1

Welcome to the Grembi Lab!

1.1 About the lab

Welcome to the lab of Dr. Jess Grembi, Assistant Professor of Pharmacology and member of the Huck Institutes of the Life Sciences at Penn State University. Our mission is to improve child and maternal health in low- and middle-income countries by understanding the mechanisms underlying environmental enteric dysfunction. Specifically, we study the microbiota that reside in the human small and large intestine, the metabolites they produce, and how these interact with the host immune system. Our focus is on improving the health of vulnerable populations from low-resource settings, particularly with respect to nutritional status. We are an interdisciplinary team and work collaboratively with microbiologists, immunologists, statisticians, epidemiologists, clinicians, and other scientists, both locally and globally. Together we tackle pressing questions related to child and maternal health, striving to make a meaningful impact in the communities we serve. To learn more about the lab, visit the lab website.

1.2 About this lab manual

This lab manual covers our communication strategy, code of conduct, laboratory safety, and best practices for reproducibility of computational workflows. It is a living document that is updated regularly.

This manual draws heavily from the lab manual of my mentor and collaborator, Dr. Jade Benjamin-Chung. We forked that repo and several of the sections remain intact given the similar coding style that I have adopted after working with Dr. Benjamin-Chung. Original author attributions are shown at the top of

each section. Contributors from the Grembi lab include Dr. Jess Grembi, Pablo Ochoa Andersen, and Nazifa Tabassum.

Feel free to draw from this manual (and please cite it if you do!).

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Chapter 2

Culture, conduct, and the co-lab

by Jess Grembi

2.1 Lab culture

We are committed to a lab culture that fosters creativity, integrity, enthusiasm, inclusivity, and rigor. As an interdisciplinary team, our differences are our strength. We aim to work with each other in a collaborative, supportive, inclusive, and open manner, and our lab space is free from discrimination and harassment.

We aspire to be a lab where everyone feels motivated to share their thoughts and ideas in a respectful and constructive way. Each of us sees things differently and comes to the table with different expertise. Even new lab members who are still learning about our research can offer valuable critical evaluation of our work and are exceptional resources to gain feedback on what is not clear in how we present our research.

We work together to share our knowledge and seek assistance when needed. This lab manual is a resource for sharing such information with each other, in particular how to get your analyses running smoothly. Please let Dr. Grembi know if you have ideas about new content to add!

2.2 Protecting human subjects

All lab members must complete CITI Biomedical Human Subjects Research (IRB) Course Stage 1 - Basic Course training and share their certificate with Jess. She will add team members to relevant Institutional Review Board protocols

prior to their start date to ensure they have permission to work with identifiable datasets.

One of the most relevant aspects of protecting human subjects in our work is maintaining confidentiality. For students supporting our data science efforts, in practice this means:

- Be sure to understand and comply with project-specific policies about where data can be saved, particularly if the data include personal identifiers.
- Do not share data with anyone without permission, including to other members of the group, who might not be on the same IRB protocol as you (check with Jess first).

Remember, data that looks like it does not contain identifiers to you might still be classified as data that requires special protection by our IRB or under HIPAA, so always proceed with caution and ask for help if you have any concerns about how to maintain study participant confidentiality.

2.3 Authorship

Team members who meet the ICMJE Definition of authorship will be included as co-authors on scientific manuscripts.

2.4 Responsible use of AI tools

This guidance draws heavily from a version shared by Dr. Kim Meier (Assistant Professor at the University of Houston)

I encourage lab members to think of large language models (LLMs) and other AI tools the way we think about calculators or Wikipedia: they can be incredibly useful, but only if you already have an understanding of what you're doing. They're best used as assistants – not as authorities or substitutes for careful thinking. This guide outlines how I expect AI tools to be used in our lab context. This guide was written with the help of ChatGPT.

2.4.1 Philosophy

AI tools can help us work more efficiently, learn more quickly, and get past sticking points – whether that's debugging a script, brainstorming how to visualize data, or wading through a dense research article. I see them as part of the broader toolkit we use to support our work, like textbooks, search engines, or discussion with labmates. But just like those other tools, the usefulness of AI depends on the thoughtfulness of the person using it. It can help you think, but it can't do the thinking for you. Think of AI tools like a lab whiteboard: a place to sketch ideas, not publish results. It will not take responsibility for errors, misinterpretations, or ethical lapses. That's still on you.

2.4.2 Helpful and encouraged uses

You're welcome to use AI tools to assist with tasks like:

- Googling stuff! Even excluding Gemini, Google search uses AI
- Getting help understanding a research paper or background topic
- Brainstorming or refining ideas (e.g., for a figure, a method, or a statistical approach)
- Getting feedback on a piece of writing you drafted
- Asking for explanations of code behavior or R functions
- Writing or revising your own code, as long as you understand what the code is doing
- Generating visualizations or suggesting ways to summarize data

It's fine to use AI to "talk things out." Just don't stop there – make sure you apply your own judgment and review anything it suggests carefully.

2.4.3 Use with caution: known limitations

AI tools often sound confident even when they're wrong. They can fabricate citations, introduce subtle bugs in code, or suggest statistical approaches that don't fit your design. Some specific risks include:

- Code suggestions might run but be logically incorrect
- Statistical advice might not match your data or assumptions
- Summaries of research papers can miss key details or invent conclusions

While AI tools are improving — for example, ChatGPT now offers a DeepResearch feature designed to reduce citation hallucinations and summarise literature — these are still limited to open-access articles only, and errors still happen. Always verify citations at the source. To use DeepResearch, you have to ask ChatGPT to use it as it won't search carefully by default. For example, you would need your prompt to be something like: "Using deep research, give me a paper that describes X"

If you don't already understand the concept or method, you probably won't be able to tell if the AI got it wrong. That's a sign you should pause and ask a labmate (or me) for help.

2.4.4 Boundaries and responsibilities

- No uploading data: Never paste data, participant responses, or sensitive information into AI tools – even in small samples. That includes data from active studies, even if it looks anonymous. If you're getting help with code, describe the structure of your data instead (e.g. "I have a $64 \times 1000 \times 120$ matrix called `motionData` representing channels x timepoints (in ms) x trials", or "I have a data frame in R where each row represents a participant and the columns include group, age, and threshold"). If you're not

sure how to describe your data without showing it, ask a labmate or bring it to our next meeting.

- No identifiable content: Avoid including names or sharing personal details with these tools, including information about yourself - whether you're getting help with writing a reference letter, drafting an email, or talking through a lab-related issue. Even if it seems harmless, AI tools are not private spaces.
- You're responsible for vetting the output: If you don't understand the code, math, or language an AI tool gives you, don't just copy and paste it into your project. Run it by someone else, or step back and learn the concept first. That's how you grow as a researcher. AI is here to help you work, not do your work for you.
- No AI-written research content: AI should never be used to generate text for a research paper, abstract, or manuscript section. You can use it to help organize your thoughts or improve the clarity of something you wrote - but the core ideas, language, and citations must be your own. Even if you're using AI to help you phrase something better, avoid copy-pasting text that it wrote for you. You'll learn a lot more (and avoid problems) by rewriting in your own words.
- This includes embedded AI tools: This policy applies not just to standalone tools like ChatGPT, but also to built-in features like Cursor, GitHub Copilot, Microsoft Editor, or Google Docs suggestions. Use the same judgment no matter where the AI shows up.

2.4.5 Other contexts have other rules

This guide is specific to our lab work. If you're working on coursework, a class presentation, a fellowship application, or a conference submission, you must follow their guidelines about AI use and authorship.

2.5 One Health Microbiome Center Co-laboratory

written by Bisanz lab members ###Background *Mission Statement:* The OHMC-CL exists to catalyze cutting edge microbiome science at PSU through facilitating independent access to cutting edge high-throughput technologies for microbiome profiling. *What the OHMC-CL is:* The OHMC-CL is collaborative space with shared equipment, resources, and protocols for microbiome characterization. The goal is for users across PSU to be able to access and use these resources to enable their research. *What the OHMC-CL is not:* The OHMC-CL is not a pay-for-service core. Typical core services are not provided. The ultimate success of any given project is the responsibility of the trainee or user. It is expected that all users will participate in scheduled training sessions and become independent users. Users who do not anticipate becoming independent users are instead encouraged to explore core services through the Huck Genomics Core or external service providers. Recommendations specific to the

project can be provided upon request. *What has the OHMC-CL enabled?* In the first year and a half of operation, users have profiled 6,140 samples from a variety of sources including human, mouse, soil, plant, livestock, poultry, and environmental sources. There have been a total of 34 users, from 22 labs, from 12 different departments. Users have performed a mixture of sample extraction, qPCR, amplicon, and metagenomic sequencing. *Why does the OHMC-CL exist?*

- Community support: There are 550 OHMC members across 42 departments, and 10 colleges at PSU. While our research interests are highly varied, many of us use the same 'omics-based approaches to interrogate our microbial communities. The OHMC-CL exists to facilitate the sharing of technical knowledge and resources to lower boundaries to using 'omics based approaches to study microbial communities.
- Equipment: The equipment required to process microbiome samples for sequencing costs hundreds of thousands of dollars. When each lab purchases their own, redundancy and waste are inevitable with capital equipment sitting unused for most of its useful lifetime. Through a variety of sources, including a generous donation from Qiagen, the OHMC-CL enables new labs, or those who are getting into the microbiome space, to use shared equipment creating significant cost savings which may be reinvested into doing science to enable publications and grants.
- Economy of scale/cost sharing: The major non-personnel costs of microbiome sequencing are derived from (i) extraction, (ii) library preparation, and (iii) sequencing. The OHMC-CL enables cost savings in each of these axes. Extraction: through allocations of extraction reagents provided by Qiagen, we can partially, or completely, reduce the major costs of extraction to only pipette tips and minor plastics disposables. Library preparation: through reaction scaling and custom protocols, the cost of library prep can be reduced 5-10x. Further, by trainees making their own libraries, significant savings are derived from labor while also providing marketable and transferable skills. Sequencing: sequencing costs are highly dependent on throughput ranging from \$3/Gb to \$500/Gb depending on sequencer and chemistry used. In effect, it is possible to obtain amplicon sequencing at a cost of ~\$4/sample (~100,000 reads/sample via NextSeq 2000 XLEAP reagents), and metagenomic sequencing at ~\$30/sample (10Gb/sample via NovaSeq X 25B reagents). To reach these costs, hundreds of samples need to be sequenced at a time; however, most users do not have the required number of samples to reach this economy of scale. The OHMC-CL provides an opportunity to pool samples across labs to create cost savings for our community.

###Becoming an OHMC-CL User *Intake Form and Meeting:* All new users must complete an intake form available here using their PSU login. The information collected in this form is used to help report usage both internally and externally. It is requested that each new project performed in the OHMC-CL has its own intake form. After reviewing the form, we will reach out to schedule a meeting with the trainee and PI to discuss project needs and to identify appropriate equipment and protocols.

Responsibilities of Trainees: 1) Attend a training session before using any OHMC-CL equipment. 2) After training, operate equipment independently and

responsibly. 3) Schedule equipment in advance and cancel reservations in a timely manner if needed. 4) Bring all necessary items and reagents with you to perform your work. Clean and reset equipment/workspaces after use. 5) Troubleshoot experiments with your PI. Ask for help via the OHMC-Co-lab Slack only after making a good-faith effort to resolve issues independently. 6) Contribute back to the co-lab by helping to train others, suggesting and testing protocol improvements, making amendments to protocols on github, and sharing tips via Slack.

Responsibilities of Supervisors: 1) Support trainees in troubleshooting and data interpretation within their home lab. 2) Reinforce expectations for independent use and respectful engagement with co-lab infrastructure. 3) Assist in evaluating whether a trainee's project is appropriate for the co-lab versus a fee-for-service option.

Communicating with other OHMC-CL Users: The preferred mode of communication for the OHMC-CL users is through a slack work group. An invite link is available [here](#). Please send all communications through this channel including requests for help troubleshooting. This will facilitate disseminating knowledge across the user base and allowing users to help other users.

Training: Researchers will be trained to operate laboratory equipment used in microbiome workflows, including automated liquid handlers for 96-well nucleic acid extraction, library preparation, and qPCR. Training covers how to set up, troubleshoot, and maintain instruments. Sessions are held every other month to support consistent onboarding and skill-building. The training schedule will be posted in the Training channel of the OHMC-CL slack group. Training is currently offered on: - High throughput DNA extraction using QIAcube HT and Powersoil reagents - Integra mini-96 pipettes - Primary PCR for amplicon sequencing - qPCR via the QIAquant 384 - Tapestation analysis of DNA/RNA samples - Metagenomic library preparation

Access: Users will be able to access the OHMC-CL during normal business hours; however, card access will not be given due to biosafety protocols.

Requesting donated reagents: We receive an annual donation which is allocated among users. Best efforts are made to ensure allocations are evenly spread among lab groups; however, users with funded projects are encouraged to use their support to purchase consumables, thus allowing donated resources to seed future funding.

Scheduling equipment: Equipment is available on a first-come first-served basis and must be reserved via the link in each equipment's slack channel. Frequent last minute cancellation will lead to a loss of scheduling privileges.

Contributing to the OHMC-CL: Since the OHMC-CL runs on donated equipment, reagents, and time, it is expected that all users will give back to the OHMC-CL. Examples of how users may contribute include: 1. Becoming a trainer: users who have become competent with specific equipment will be in-